

Clinical Belief State: A Persistent Patient Representation for Longitudinal Medical Vision-Language Models

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Abstract—Longitudinal chest radiography is a sequential clinical process in which radiologists interpret each new examination by integrating current imaging findings with prior studies, previous reports, temporal disease progression, and accumulated clinical knowledge. We propose sequential patient-state estimation under partial observability as a principled computational formulation for longitudinal radiology, in which a hidden clinical state is recursively inferred from incomplete, multimodal, irregularly sampled, and occasionally contradictory observations, and report generation is treated as a downstream consequence of this inference rather than the primary computational objective. Implementation analyses of representative longitudinal medical vision-language models (VLMs)—including CXRMate, CXRMate-2, LLaVA-Med, InternVL, and OpenFlamingo—indicate that existing systems instead treat history as contextual input: prior studies are concatenated into prompts, retrieved, or re-encoded independently at each inference, leaving patient state implicit rather than explicitly estimated. This paper operationalizes the patient-state estimation formulation through the Clinical Belief State (CBS), a structured variational recurrent state-space implementation that separates deterministic disease trajectory memory from a stochastic latent component, allowing evidence to be recursively integrated over time rather than repeatedly reconstructed from raw historical inputs; we treat this stochastic component as a learned representation whose calibration against clinically defined uncertainty is empirically evaluated, rather than assuming by construction that it represents clinical uncertainty. CBS instantiates this formulation using a recursive belief update module positioned between multimodal observation encoding and report generation, trained with a report generation objective together with belief consistency and KL-regularization terms in the recurrent state-space model tradition. The proposed framework will be evaluated on MIMIC-CXR, MI-CXR, OpenI, and associated longitudinal benchmarks, using conventional report-generation metrics together with evaluations of temporal consistency, uncertainty calibration, belief persistence, contradiction handling, and clinical trajectory modeling—the latter grounded in CheXTemporal-derived progression labels—framed as tests of the underlying patient-state estimation hypothesis rather than as a direct architectural comparison.

Index Terms—medical vision-language models, longitudinal chest X-ray report generation, clinical belief state, Bayesian filtering, recurrent state-space models, medical AI, temporal reasoning, clinical memory

I. INTRODUCTION

A. Clinical Motivation

Artificial intelligence has rapidly transformed automated medical image interpretation through the development of large-scale vision-language models capable of generating

clinically coherent radiology reports directly from diagnostic images. Advances in multimodal representation learning, transformer-based language modeling, and contrastive pre-training have significantly improved report quality, enabling models to describe complex imaging findings with increasing linguistic fluency and diagnostic accuracy.

Despite these advances, longitudinal radiology differs in important ways from single-image interpretation, and most existing medical vision-language models remain largely cross-sectional: they assume that each examination can be interpreted independently using only the current image or a limited amount of retrieved historical information. In practice, radiologists do not simply interpret the current image in isolation. They sequentially integrate current imaging, previous studies, prior reports, clinical history, elapsed time, and their own uncertainty to continuously revise their understanding of the patient’s condition.

This integration process is directed at a specific set of clinical questions: has disease progressed? Is this finding stable? Has it resolved? Is this a recurrence? Is the prior evidence still valid given what is seen today? These questions concern the patient’s evolving condition rather than any single isolated observation, and answering them requires an internal representation of the patient that persists and is revised across encounters, not merely a fresh interpretation of each new image.

B. Problem Formulation

These clinical observations motivate a specific computational framing, which we investigate rather than assert as given. We propose sequential patient-state estimation under partial observability as a principled formulation for longitudinal radiology, rather than treating the problem as image or text generation conditioned on additional context.

We stress that the logical chain here runs in one direction, not by mutual implication: the clinical workflow described in Section I-A *motivates* the computational formulation introduced below; the formulation in turn motivates a specific class of implementations, including the one proposed in this paper; and that implementation is then subjected to empirical test (Section VI). Each step is a proposal to be evaluated, not a deduction from the previous one.

We also note that the latent patient state introduced below is a computational abstraction adopted for modeling purposes. We do not assume it corresponds to a uniquely identifiable biological or physiological state; it is a modeling device whose usefulness is judged by whether it improves longitudinal reasoning on the evaluation criteria in Section VI, not by any claim of biological correspondence.

Under this framing, the patient’s true clinical state is hidden and never directly observed. The available observations—images, reports, clinical metadata, and the elapsed time between encounters—are incomplete, multimodal, irregularly sampled, and at times contradictory. The computational objective is therefore to estimate the evolving latent patient state from this sequence of partial observations, rather than to merely generate the next report from the most recently available context. Report generation is naturally positioned as a downstream task, conditioned on the estimated patient state, rather than as the primary object of modeling.

C. Gap in Existing Literature

Existing longitudinal medical vision-language approaches can be broadly summarized into four groups: history-concatenation methods that append prior reports or images directly into the model input; retrieval-based methods that retrieve relevant historical studies at inference time; temporal fusion methods that combine current and historical multimodal features through attention or fusion layers; and longitudinal report generation systems that combine several of these mechanisms end to end.

These methods primarily treat historical information as additional context supplied to a report-generation model. They do not explicitly formulate the problem as recursive estimation of an evolving patient state: the trajectory of the patient’s condition is not represented as an explicit computational object, but is instead implicit in however the model happens to combine the available context. This is not to say that these methods are incorrect—they demonstrably improve report quality over cross-sectional baselines. Rather, the patient state they rely on remains implicit rather than explicitly formulated, estimated, or evaluated as such.

D. Our Position

Building on this formulation, we propose modeling longitudinal radiology as recursive inference over an evolving latent patient state, and we instantiate this proposal with the Clinical Belief State (CBS).

We emphasize that CBS is one implementation of the patient-state estimation formulation, not the formulation itself, and not a new probabilistic framework in its own right. Nor is patient-state estimation itself a new idea: Deep Kalman Filters [11] and the associated Structured Inference Networks [12] learn a recursive, uncertainty-aware latent representation of a patient that evolves over time from longitudinal electronic health records, the Deep Physiological State Space Model [13] extends this line to intervention-augmented forecasting of future clinical observations, and Nakamura et

al. [28] extend the underlying deep state-space approach again in 2023, applying it to unsupervised clustering of cancer patients’ latent trajectories from longitudinal laboratory time series for prognosis visualization. These are the direct conceptual ancestors of the formulation used here, developed for structured or tabular EHR data beginning roughly a decade before this proposal and continuing into recent years, rather than a single decade-old idea left unrevisited. CBS operationalizes the same estimation problem through a structured latent dynamical model in the recurrent state-space model (RSSM) tradition [3], separating a deterministic disease trajectory memory from a stochastic, uncertainty-aware clinical belief component. The delta we claim over this prior art is narrow and specific, not architectural: (i) a heterogeneous, multimodal observation model combining imaging, free-text reports, and clinical metadata rather than structured EHR fields alone; (ii) an LLM-based generative decoder for free-text report generation conditioned on the estimated state; and (iii) a dedicated, pre-registered stress-test protocol (Section VI-I) targeting partial observability, contradiction, and long-horizon drift specifically in longitudinal chest radiography. The central claim of this paper is therefore not that CBS introduces a new class of probabilistic model, but that extending this established estimation paradigm to multimodal, image-and-text longitudinal radiology, and testing it under conditions designed to isolate the effect of explicit state estimation from general recurrent modeling capacity, provides a principled computational formulation that we hypothesize better supports longitudinal reasoning than history concatenation or retrieval-based context conditioning.

We further note that the clinical problem motivating this work is not unaddressed in practice: deployed commercial systems such as Harrison.Rad [22], and academic systems such as MAIRA-2 [17] and BioViL-T [15], already condition report generation on prior studies at scale, via context-conditioning rather than explicit recursive state estimation. We do not position CBS as a competing report generator against these systems—at MIMIC-CXR scale, an academic project cannot match a system trained on millions of proprietary image-report pairs on aggregate report-quality metrics, and we do not attempt to. We instead position CBS as a controlled, mechanistic study of whether explicit, uncertainty-aware persistent state estimation offers advantages over context-conditioning specifically under partial observability, holding data and (where feasible) model capacity fixed—a question that remains open regardless of how strong any single deployed system’s aggregate report quality is.

Competing formulations. Retrieval-based, temporal-fusion, and context-conditioned approaches represent alternative computational formulations of longitudinal reasoning, and each has demonstrated measurable benefits over cross-sectional baselines. This work does not claim that patient-state estimation is the only valid formulation, nor that it strictly dominates these alternatives. Rather, it investigates whether explicit recursive state estimation provides a useful inductive bias under clinically relevant conditions such as partial observability, irregular

sampling, and evolving or contradictory patient trajectories, where an explicit persistent state may offer advantages that context concatenation or retrieval do not.

Falsifiability. This hypothesis would be challenged—and we would consider it not supported—if explicit patient-state estimation failed to provide measurable advantages over retrieval- or context-conditioned baselines under stress-test conditions such as incomplete, contradictory, or long-horizon clinical histories (described in Section VI). Under that outcome, the more parsimonious conclusion would be that observed gains from longitudinal architectures are attributable to general recurrent modeling capacity rather than to patient-state semantics specifically, a possibility we return to as a threat to validity in Section VI.

A narrower, more defensible version of the thesis. Stripped to its most testable form, independent of any claim about clinical belief or biological correspondence, the question this paper answers is: *does a fixed-dimensional, recursively updated latent representation retain clinically useful longitudinal information better than repeatedly reconstructing or retrieving history, under controlled comparisons that hold parameter count and recurrent capacity fixed?* We do not need to establish that CBS recreates how radiologists reason, that Z_t corresponds to a real physiological state, or that CBS broadly outperforms modern context-conditioned systems at scale (Section I-D) to answer this question; we need only establish CBS > history retrieval under the specific partial-observability stress conditions in Section VI, with capacity controlled for. This narrower framing is what the falsifiability criterion above and the evaluation design in Section VI are built to test, and is the claim we consider this proposal accountable to.

II. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

A. Background

Medical vision-language models (VLMs) have demonstrated remarkable progress in automatic chest X-ray report generation through advances in multimodal representation learning and large language models. Recent longitudinal VLMs extend these capabilities by incorporating previous radiographs, historical reports, temporal information, or multimodal retrieval to improve report quality. These systems recognize that radiological interpretation is inherently longitudinal and that previous examinations provide clinically valuable context for understanding disease progression.

Despite these advances, the computational paradigm underlying existing longitudinal medical VLMs remains largely based on historical context reconstruction. At each inference, prior reports are concatenated into prompts, previous images are retrieved and re-encoded, or historical multimodal features are recomputed to condition report generation. The model therefore reconstructs historical evidence repeatedly rather than maintaining an explicit representation of the patient’s evolving health state.

Comprehensive implementation audits conducted for this research support this observation across representative open-

source architectures. CXRMate and CXRMate-2 reconstruct historical reports and images during each inference without maintaining a persistent patient representation. LLaVA-Med and InternVL remain stateless beyond the current context window. OpenFlamingo stores temporary visual features only during inference, while Memory Layers at Scale introduces persistent parametric key-value storage but represents global model knowledge rather than patient-specific state. DreamerV3 demonstrates recursively updated latent state estimation through recurrent state-space modeling but is designed for sequential world modeling rather than longitudinal clinical reasoning.

Collectively, these systems illustrate that existing architectures implement various forms of storage, retrieval, prompt conditioning, or latent sequential memory, yet none maintain an explicit, patient-specific, recursively updated computational representation of clinical belief.

B. Problem Statement

We model clinical reasoning as a process of belief revision. Radiologists do not reconstruct every previous examination independently whenever a new study is acquired. Instead, they maintain an evolving internal understanding of the patient’s condition and continuously update this understanding as new clinical evidence becomes available. Current observations modify previous beliefs regarding disease progression, stability, improvement, or deterioration.

Current longitudinal medical vision-language models do not explicitly model this process. Instead, they treat longitudinal reasoning primarily as a retrieval problem:

Historical Studies → Retrieve → Re-encode → Generate Report
(1)

whereas clinical reasoning is more naturally described as

Previous Belief → New Evidence → Updated Belief → Report.
(2)

The absence of a persistent computational representation of patient state creates several limitations. First, patient knowledge must be reconstructed repeatedly from raw historical observations, leading to redundant computation and dependence on complete historical retrieval. Second, existing architectures lack an explicit mechanism for recursively integrating new evidence into an evolving patient representation; disease progression is therefore inferred implicitly through prompt conditioning or multimodal attention rather than through formal state estimation. Third, uncertainty regarding latent patient condition is not explicitly represented—although language models generate probabilistic token distributions, these probabilities describe linguistic uncertainty rather than uncertainty about the patient’s underlying clinical state. Fourth, contradictory longitudinal observations are handled implicitly during language generation rather than through principled belief revision, so existing systems provide no explicit computational mechanism for updating patient state when new observations conflict with previous expectations.

These limitations suggest that current longitudinal medical VLMs model clinical *history* rather than clinical *belief*.

C. Research Gap

The literature and repository analyses reveal three distinct categories of memory used in contemporary neural architectures.

The first consists of *transient inference memory*, including transformer key-value caches, prompt histories, and temporary visual feature representations. These mechanisms persist only within a single inference session and are discarded afterward.

The second consists of *persistent parametric memory*, exemplified by Memory Layers at Scale, where large trainable key-value stores are learned through gradient descent. These memories represent global model knowledge and are shared across all inputs rather than maintaining individualized patient representations.

The third consists of *persistent latent sequential state*, as implemented in recurrent state-space models such as DreamerV3. These architectures recursively estimate latent environment states over time but were developed for sequential decision-making and world modeling rather than longitudinal clinical reasoning.

None of these approaches maintain a persistent, patient-specific, uncertainty-aware clinical belief state that recursively evolves across successive medical encounters. Consequently, a significant gap exists between the computational memory mechanisms used by current neural architectures and the belief revision process employed during longitudinal clinical reasoning.

D. Proposed Research

This research proposes a computational formulation of longitudinal radiology as sequential patient-state estimation under partial observability, and instantiates this formulation through the Clinical Belief State (CBS), a persistent latent representation that summarizes the current estimated health state of an individual patient by recursively integrating multimodal clinical observations over time.

Unlike retrieval-based longitudinal models, CBS does not repeatedly reconstruct patient history. Instead, each new clinical encounter updates the previous patient belief through recursive state estimation in the spirit of Bayesian filtering and recurrent state-space modeling. The proposed Clinical Belief State consists of two complementary components: (1) a deterministic disease trajectory memory that captures long-term patient evolution, and (2) a stochastic latent clinical belief that explicitly represents uncertainty regarding the patient’s current condition. Together, these components provide a neural approximation of the posterior distribution over latent patient health conditioned on all available clinical observations. CBS is best understood as one implementation of this formulation—a structured latent dynamical model belonging to the broader RSSM / variational state-space family—rather than as a new probabilistic modeling paradigm. Rather than treating report

generation as the primary computational objective, the proposed framework formulates longitudinal report generation as a downstream consequence of recursive patient-state estimation.

E. Research Question

The central research question addressed in this work is:

Can a persistent, recursively updated Clinical Belief State improve longitudinal medical vision-language reasoning by explicitly modeling evolving patient state rather than repeatedly reconstructing historical context?

This central question is supported by the following sub-questions:

- How should patient health be represented as a persistent latent computational state suitable for longitudinal medical reasoning?
- How should multimodal clinical observations recursively update this patient belief over time?
- Can deterministic disease trajectory memory and stochastic uncertainty-aware latent belief jointly improve longitudinal patient modeling?
- Does explicit patient-state estimation improve longitudinal report generation compared with retrieval-and-reencoding approaches?
- Can recursive belief updating provide more consistent modeling of disease progression, contradiction resolution, and uncertainty than existing longitudinal medical vision-language models?

F. Research Hypothesis

We hypothesize that longitudinal clinical reasoning can be fruitfully modeled as a recursive state-estimation problem rather than solely as a historical retrieval problem. Accordingly, we predict that maintaining a persistent Clinical Belief State will: (i) improve temporal consistency across longitudinal reports; (ii) improve the representation of disease progression relative to retrieval-based history encoding; (iii) enable principled uncertainty-aware patient modeling; (iv) facilitate explicit belief revision when new observations contradict previous patient states; and (v) outperform retrieval-based longitudinal architectures in modeling evolving patient health under the stress-test conditions described in Section VI. As noted above, this hypothesis is treated as falsifiable: a failure to observe these advantages, particularly under incomplete or contradictory histories, would count as evidence against it.

G. Research Questions (Formal)

- **RQ1:** Does a persistent Clinical Belief State improve longitudinal report generation?
- **RQ2:** Does recursive belief updating improve temporal consistency?
- **RQ3:** Can CBS better model disease progression than retrieval-based history encoding?
- **RQ4:** Does explicit uncertainty modeling improve calibration?

- **RQ5:** Does CBS produce interpretable latent representations of patient evolution?

H. Research Objectives

- 1) Formally define the Clinical Belief State and its recursive update equations.
- 2) Design a belief-conditioned report generation architecture integrating CBS into a longitudinal medical VLM pipeline.
- 3) Develop a training objective combining report generation, belief consistency, and latent regularization losses.
- 4) Empirically evaluate CBS against retrieval-based longitudinal baselines using report-quality, clinical-accuracy, longitudinal-reasoning, and uncertainty metrics.
- 5) Analyze the interpretability and stability of the learned latent belief trajectories.

III. RELATED WORK

We organize related work into five areas: longitudinal report generation, clinical trajectory modeling, latent dynamical systems, Bayesian approaches to diagnosis, and patient-state estimation. CBS sits at the intersection of these areas, and is best understood by considering what each area contributes and what it leaves unaddressed.

A. Longitudinal Report Generation

Most medical VLMs, including general-purpose multimodal architectures adapted to radiology, generate reports from a single image without explicit modeling of patient history. Longitudinal systems that condition on prior studies include CXRMate [1] and CXRMate-2 [2]¹, BioViL-T [15], a contrastive image-report pretraining model built explicitly for temporal representation learning; MAIRA-1 and MAIRA-2 [16], [17], grounded report generation models that condition on prior studies and now serve as standard comparison points in this sub-area; HERGen [18], a history-encoding report generation baseline; PriorRG [19], a 2026 longitudinal report generator; TIM [29], a 2026 temporal-decoupling, iterative mutual-refinement longitudinal report generator, mentioned here as a further recent context-conditioned baseline in the same wave as PriorRG and MLRG; MLRG [20], which introduces a tokenized-absence-encoding mechanism specifically to handle missing prior reports or indications, directly relevant to the missingness problem in our own observation model (Section IV-C); and RA-RRG [21], a retrieval-augmented, LLM-fine-tuning-free approach targeting the compute-efficiency objection that recurrent, variational architectures such as CBS must also confront. General-purpose multimodal backbones

¹CXRMate-2 is a distinct, more recent system from the same research group, integrating structured multimodal conditioning with a reinforcement-learning composite reward for semantic alignment with radiologist reports, reporting gains of approximately 11.2% and 24.4% over MedGemma 1.5 on GREEN and RadGraph-XL respectively across MIMIC-CXR, CheXpert Plus, and ReXGradient. A companion qualitative retrospective pilot study (Cooper et al.) additionally reports a blinded evaluation of CXRMate-2 by three consultant radiologists across 120 MIMIC-CXR studies; we discuss the implication of this precedent for our own planned qualitative evaluation in Section VI-K.

such as LLaVA-Med [4], InternVL [8], and OpenFlamingo [9] underlie many of these systems. Repository audits conducted for this proposal indicate that, mechanistically, these systems reconstruct historical context—via transformer key-value caches, prompt histories, or temporary visual feature representations—at every inference call rather than maintaining a persistent, recursively updated patient-level state; this mechanism persists only within a single inference session and is discarded afterward, requiring historical context to be reconstructed at every subsequent encounter.

Beyond academic systems, deployed commercial models now address the same clinical motivation at a scale no public-data academic project can match: Harrison.Rad [22], trained on approximately six million image-report pairs and deployed across more than 1,000 sites, explicitly compares a current study against a prior to describe interval change in natural radiologist prose, via domain-adapted LLM and contrastive-encoder context-conditioning rather than explicit recursive state estimation. We discuss the implications of this for how CBS’s contribution should be framed in Section VIII.

B. Clinical Trajectory Modeling

A separate line of work models disease trajectories from structured or semi-structured longitudinal clinical data (e.g., laboratory values, vitals, coded diagnoses) using sequence models such as RNNs and Transformers to predict future clinical events or trajectories. These approaches demonstrate the clinical value of explicit trajectory modeling but are typically applied to structured tabular or time-series data rather than to multimodal imaging and free-text reports, and generally do not maintain an explicit latent belief with calibrated uncertainty.

C. Latent Dynamical Systems

State estimation has long been studied in control theory, robotics, and reinforcement learning. Bayesian filters recursively update a belief distribution over hidden states given sequential observations, and recurrent state-space models (RSSMs) combine a deterministic recurrent state with a stochastic latent variable, using prior prediction, posterior correction, and KL-balanced training. DreamerV3 [3] is a representative implementation of this family, developed for sequential decision-making and world modeling. These techniques provide the mathematical machinery for recursive belief updating but have not been systematically adapted to longitudinal medical imaging as an explicit patient-state representation.

D. Bayesian Approaches to Diagnosis

Bayesian networks and probabilistic graphical models have a long history in computer-aided diagnosis, representing diseases and findings as random variables linked by conditional dependencies and supporting principled uncertainty propagation. These approaches formalize diagnostic uncertainty but are typically static, cross-sectional, and reliant on hand-specified structure, rather than learned end-to-end from raw multimodal longitudinal data.

E. Patient-State Estimation

A specific body of work explicitly frames clinical modeling as state estimation, treating the patient’s underlying condition as a hidden variable to be inferred from sequential, partially observed clinical evidence. Deep Kalman Filters [11] and Structured Inference Networks [12] learn exactly such a recursive, uncertainty-aware patient representation from longitudinal EHR data, and the Deep Physiological State Space Model [13] extends this to intervention-augmented forecasting. These are the direct conceptual ancestors of the formulation used in this paper (Section I-D), developed for structured or tabular clinical time series beginning roughly a decade before this proposal. More recently, Nakamura et al. [28] (2023) proposed a deep state-space framework for unsupervised patient latent-state clustering from longitudinal laboratory time series, extending this line within the last several years rather than leaving it only in its original 2015–2019 form. We note this narrows the *temporal* novelty gap (revisited as recently as 2023, not left dormant since 2015–2019) more than it narrows the *modality* gap this proposal actually rests its claim on, since Nakamura et al. address unsupervised trajectory clustering over tabular lab values rather than generative, report-conditioned state estimation over multimodal image-and-text streams; we therefore narrow our own novelty framing accordingly (Section I-D): the open problem is not that patient-state estimation has been dormant for a decade, but that it has not been extended to heterogeneous, partially-observed image-and-text streams for generative report conditioning, which remains the specific gap this proposal addresses. These works are engaged with here at the depth their role as primary prior art warrants rather than as a single dismissive mention. EHRXDiff [14] is a further recent, non-RSSM instantiation of a related idea, using latent diffusion conditioned on a prior chest X-ray and subsequent EHR events to predict future radiographic appearance; it targets a generative-forecasting objective rather than recursive belief estimation for report generation, but is conceptually adjacent and is discussed here as a parallel design point. Based on our literature and repository analyses, none of this line has previously been instantiated for multimodal (image-and-text) longitudinal chest radiography using a structured variational recurrent state-space model with an LLM-based report decoder.

F. A Taxonomy of Neural Memory Mechanisms

Across these areas, repository-level audits conducted for this proposal identify three distinct categories of memory in contemporary neural architectures, none of which constitutes a persistent, patient-specific, uncertainty-aware clinical belief state.

Transient inference memory includes transformer key-value caches, prompt histories, and temporary visual feature representations, as used in CXRMate, LLaVA-Med, InternVL, and OpenFlamingo. These mechanisms persist only within a single inference session and are discarded afterward.

Persistent parametric memory, exemplified by Memory Layers at Scale [10], employs large trainable key-value stores

learned through gradient descent. These memories represent global model knowledge shared across all inputs rather than individualized, patient-specific representations, and are therefore not suited to tracking a single patient’s evolving state.

Persistent latent sequential state, as implemented in recurrent state-space models such as DreamerV3, and in the Deep Kalman Filter / Deep Markov Model line [11]–[13], recursively estimates a latent state over time using deterministic and stochastic components. This is architecturally and conceptually closest to the proposed instantiation; DreamerV3 was developed for sequential decision-making and world modeling, while the Deep Kalman Filter line was developed for structured EHR time series, and neither has been adapted to multimodal (image-and-text) longitudinal clinical belief.

Positioned within this landscape, CBS does not introduce a new memory mechanism or a new probabilistic modeling family. Rather, it combines the recursive belief updating of the Deep Kalman Filter / RSSM literature with a multimodal, LLM-decoded observation model and a dedicated clinical stress-test protocol, applied, based on our literature and repository analyses, for the first time to multimodal longitudinal chest radiography.

G. Recent Longitudinal Chest X-Ray Benchmarks

Two recent benchmarks are directly relevant to the evaluation design in Section VI and are used there rather than only cited here. MI-CXR [23] is a benchmark of five-visit CXR timelines testing temporal event localization, interval-wise change reasoning, and global trajectory summarization; it reports that fourteen current state-of-the-art VLMs score only approximately 29% on its tasks, barely above the 20% chance rate for its five-way questions, providing independent, recent evidence that current longitudinal VLMs do not perform genuine longitudinal reasoning and directly supporting the motivation in Section I. CheXTemporal [26] provides a five-class progression taxonomy (including new and resolved findings) with spatial-temporal alignment across paired prior-current studies spanning MIMIC-CXR, CheXpert, and ReX-Gradient; we use its labels as ground truth for the Disease Progression Accuracy and Longitudinal State Tracking metrics in Section VI, replacing the previously unspecified notion of “known disease trajectories.”

MS-CXR-T [27], introduced in the same paper as BioViL-T [15] (Section III-A), already provides an incumbent three-class progression label set (improving/stable/worsening) derived from Chest ImaGenome for 1,326 studies across five findings, and has been used as ground truth in prior longitudinal-reasoning evaluations. We use CheXTemporal rather than treat MS-CXR-T as an unaddressed alternative, following the comparison CheXTemporal’s own introducing work draws against MS-CXR-T: a finer five-class taxonomy, including new and resolved findings that MS-CXR-T’s three classes do not distinguish, with spatial grounding across a broader multi-source pool. Where CheXTemporal coverage is unavailable for a given trajectory, we fall back to MS-CXR-T labels rather than omitting the comparison, so the two

label sets are used complementarily rather than one silently supplanting the other. We have not yet verified the overlap between CheXTemporal’s coverage and our ≥ 5 -visit MIMIC-CXR long-horizon subset (Section VI-B) specifically; this is scheduled as an explicit check during the dataset-audit phase (Table ??, Months 3–4), since it determines what fraction of the long-horizon stress test receives CheXTemporal ground truth versus requiring the MS-CXR-T fallback.

H. Uncertainty Quantification for Medical VLMs

Uncertainty quantification for medical VLMs and report generation is an active area, including semantic-entropy-based methods and verbalized-confidence calibration for clinical VQA. These methods are almost exclusively *post-hoc* and inference-time: they estimate uncertainty by sampling multiple generations from a fixed model and measuring their disagreement, or by asking the model to report a confidence score, without any dedicated latent variable or training-time uncertainty objective. CBS’s uncertainty is structural and trained: Z_t is a latent variable with an explicit KL-regularized prior–posterior objective (Sections IV-E and IV-H), estimating uncertainty over the patient’s underlying condition rather than over token-level generation. We distinguish these two notions of uncertainty explicitly because they answer different questions—token-level generation confidence versus confidence about the underlying clinical state—and because post-hoc methods provide natural, currently-missing baselines for the calibration metrics in Section VI-F-4, which we adopt accordingly.

IV. PROBLEM FORMULATION AND THE CLINICAL BELIEF STATE

We first state the patient-state estimation problem in general form, independent of any specific implementation, and then describe how the Clinical Belief State (CBS) instantiates it as a structured variational recurrent state-space model. The mathematical model in this section is unchanged from our original formulation; what changes is its framing: the general estimation problem is stated first, and CBS is presented explicitly as one implementation of it.

A. Problem Formulation

Let $E_{1:t}$ denote all clinical observations available up to and including encounter t . Each encounter contributes an observation

$$E_t = (I_t, R_t, M_t, \Delta t) \quad (3)$$

where I_t is the current image, R_t is the report associated with encounter t (when available), M_t is auxiliary clinical metadata, and Δt is the elapsed time since the previous encounter. Consistent with the partial-observability framing in Section I, I_t , R_t , and M_t may be incomplete, irregularly sampled, or occasionally contradictory across encounters.

Let s_t denote the latent patient state at encounter t —an unobserved variable summarizing the patient’s true underlying

clinical condition. The central inference problem addressed by this work is

$$p(s_t \mid E_{1:t}), \quad (4)$$

i.e., recursively estimating the posterior distribution over the hidden patient state given all observations to date. Report generation is then formulated as a downstream task, conditioned on the estimated state and the observation history:

$$p(r_t \mid s_t, E_{1:t}). \quad (5)$$

We emphasize that the state-estimation posterior $p(s_t \mid E_{1:t})$ is the central computational object of interest in this work; the report-generation term $p(r_t \mid s_t, E_{1:t})$ is a consequence of it, not the primary modeling target. Any architecture that produces a persistent, recursively updated approximation to $p(s_t \mid E_{1:t})$ constitutes a valid instantiation of this formulation. The remainder of this section describes CBS, a specific implementation of s_t as a structured latent dynamical model.

B. CBS as an Implementation of the Formulation

CBS operationalizes s_t by decomposing it into a deterministic component and a stochastic component, and instantiates the observation E_t used above as O_t below, following the recurrent-state-space-model (RSSM) family. We describe CBS as a structured variational recurrent state-space model implementing patient-state estimation, rather than as a new probabilistic modeling family in its own right; the underlying inference machinery belongs to the established RSSM / variational state-space literature, and the contribution lies in adapting and applying it to longitudinal, multimodal clinical data with explicit clinical state semantics.

C. Observation Model

At encounter t , the observation used by CBS is defined as

$$O_t = (I_t, R_{t-1}, M_t, \Delta t) \quad (6)$$

where I_t is the current image, R_{t-1} is the previous report, M_t is available metadata, and Δt is the temporal interval since the previous encounter. O_t is the CBS-specific instantiation of the general observation E_t introduced above.

1) *Handling a Heterogeneous, Partially Missing Observation*: Unlike the single-observation-type setting assumed in standard RSSM presentations, O_t is heterogeneous and, in realistic MIMIC-CXR trajectories, frequently partially missing (e.g., an image without a paired report, or a report without a usable prior image). We handle this in the encoder rather than the objective: each component of O_t is encoded independently, missing components are represented by a learned mask-conditioned embedding rather than omitted or zero-filled, and the posterior network $q(Z_t \mid H_t, O_t)$ is trained to marginalize over the observed subset by conditioning on the same mask indicator used at encoding time, so that $q(Z_t \mid H_t, O_t^{\text{obs}})$ is well-defined for any observed subset $O_t^{\text{obs}} \subseteq O_t$. This is closely related to, and directly informed by, the tokenized-absence-encoding mechanism proposed by MLRG [20] for missing prior reports and indications; we adopt

the same masked-token principle inside the CBS posterior network rather than proposing an alternative, and note this explicitly as an adopted mechanism rather than an independent contribution. The report-generation term $\mathcal{L}_{\text{report}}$ and the KL term in Section IV-H are computed only over the observed components of O_t at each step, which keeps the training objective well-posed on trajectories with missing modalities.

Proof sketch of the masked ELBO. Rather than deferring this entirely, we sketch the approximation here. For fully observed O_t , the standard RSSM step contributes the evidence lower bound $\log p(O_t | H_t) \geq \mathbb{E}_{q(Z_t | H_t, O_t)}[\log p(O_t | H_t, Z_t)] - D_{KL}(q(Z_t | H_t, O_t) \| p(Z_t | H_t))$. When only a subset $O_t^{\text{obs}} \subseteq O_t$ is observed, we replace $q(Z_t | H_t, O_t)$ with $q(Z_t | H_t, O_t^{\text{obs}})$, using the mask-conditioned encoding described above, and the corresponding bound becomes $\log p(O_t^{\text{obs}} | H_t) \geq \mathbb{E}_{q(Z_t | H_t, O_t^{\text{obs}})}[\log p(O_t^{\text{obs}} | H_t, Z_t)] - D_{KL}(q(Z_t | H_t, O_t^{\text{obs}}) \| p(Z_t | H_t))$, i.e., a valid ELBO on the marginal likelihood of the *observed* components only, obtained by treating the missing components as simply absent from the likelihood term rather than imputed. The approximation gap relative to the fully-observed bound is the information lost by never modeling $p(O_t^{\text{missing}} | H_t, Z_t)$ at all; because we do not attempt to reconstruct missing components (Section IV-C.1 handles missingness at the encoder, not via imputation), this gap is not directly estimable from the training objective alone. We treat closing this gap—e.g., bounding it via an auxiliary imputation model trained only for evaluation, not for the main objective—as a concrete, still-open piece of the full derivation to be completed during implementation rather than at proposal stage, and report it explicitly here as a proof *sketch* rather than a complete derivation.

D. Belief State

The Clinical Belief State instantiates the latent patient state s_t by combining a deterministic component H_t and a stochastic latent component Z_t :

$$B_t = (H_t, Z_t) \quad (7)$$

H_t captures deterministic disease trajectory memory, while Z_t is a stochastic latent component whose role as “uncertainty-aware clinical belief” is a hypothesis to be empirically tested (Section VI-F.4), not an assumption built into the model by construction: a stochastic latent variable being present does not by itself establish that its variance corresponds to clinical uncertainty about the patient’s condition, rather than simply whatever residual variability helps the decoder predict report tokens. We evaluate this distinction explicitly rather than presupposing it (Section III-H).

E. Recursive Update

Given the previous belief state B_{t-1} and new observation O_t , the belief is recursively updated as

$$H_t = f_{\theta}(H_{t-1}, Z_{t-1}, O_t) \quad (8)$$

$$p(Z_t | H_t) = \text{Prior}_{\theta}(H_t) \quad (9)$$

$$q(Z_t | H_t, O_t) = \text{Posterior}_{\theta}(H_t, O_t) \quad (10)$$

The prior predicts expected patient evolution, while the posterior incorporates the current observation to correct that prediction.

F. Identifiability and Latent Geometry

Recurrent latent-variable models of this family are known to be susceptible to posterior collapse and to non-identifiable latent geometry: the learned (H_t, Z_t) space is defined only up to invertible reparameterizations (e.g., rotation or permutation of latent dimensions) that leave the likelihood unchanged, so apparent structure in the latent space is not automatically clinically meaningful. We treat this as a property to be tested rather than assumed: Section VI-G replaces purely visual latent-trajectory inspection (UMAP/t-SNE) with a quantitative clustering metric (silhouette score and k NN purity against CheXTemporal- and CheXpert-derived disease labels), evaluated with statistical testing across independently trained seeds, so that claims of disease-specific latent structure are not dependent on a single, potentially cherry-picked visualization or a single training run.

G. Report Generation

The report at encounter t is generated conditioned on the updated belief state:

$$R_t \sim p_{\theta}(R_t | H_t, Z_t) \quad (11)$$

H. Training Objective

The overall loss combines a report generation term, a belief consistency term, and a KL regularization term between prior and posterior:

$$\mathcal{L} = \mathcal{L}_{\text{report}} + \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 D_{KL}(q(Z_t | \cdot) \| p(Z_t | \cdot)) \quad (12)$$

where $\mathcal{L}_{\text{report}} = -\log p_{\theta}(R_t | H_t, Z_t)$ is the standard report-generation negative log-likelihood, restricted at each step to the observed subset of O_t (Section IV-C), and the KL term is the usual RSSM prior-posterior divergence, computed with free-bits or KL balancing to mitigate posterior collapse (Table III).

$\mathcal{L}_{\text{consistency}}$ was left undefined in earlier drafts of this proposal; we now define it explicitly as a one-step-ahead prior-prediction penalty, separate from the KL term above:

$$\mathcal{L}_{\text{consistency}} = \mathbb{E} [D_{KL}(q(Z_t | H_t, O_t) \| \text{Prior}_{\theta}(H_{t-1}, Z_{t-1}))] \quad (13)$$

i.e., the divergence between the realized posterior at encounter t and the prior’s prediction of Z_t made *before* observing O_t (using H_{t-1}, Z_{t-1} rather than H_t). This differs from the standard KL term, which regularizes the posterior against the prior conditioned on the current H_t (i.e., after the deterministic state has already incorporated O_t via f_{θ}); $\mathcal{L}_{\text{consistency}}$ instead penalizes the model’s ability to predict its own future belief update before seeing new evidence, and is the term most directly responsible for encouraging temporally smooth, non-oscillatory belief trajectories (Section VI-G). If, during implementation, this term is found to be redundant with the standard KL term for a given choice of H_t -conditioning, we will drop it and report the ablation (Ablation C) rather than retain an unnecessary loss term for its own sake.

V. PROPOSED ARCHITECTURE

The proposed system operationalizes the patient-state estimation formulation of Section IV by introducing a recursive belief update module positioned between multimodal observation encoding and report generation. Candidate implementation components are summarized in Table I.

TABLE I
CANDIDATE ARCHITECTURAL COMPONENTS

Module	Candidate Implementation
Vision encoder	ViT / BiomedCLIP / DINOv2
Report encoder	BioClinicalBERT
Disease memory	GRU / Mamba / State Space Model
Posterior network	MLP
Prior network	MLP
Report generator	LLaMA-family medical LLM

Unless otherwise stated, all baseline models will use the same backbone vision encoder and language model where feasible, in order to isolate the effect of the Clinical Belief State.

VI. EXPERIMENTAL DESIGN AND EVALUATION

A. Objective

The primary objective of the experimental program is to test the patient-state estimation hypothesis introduced in Sections I and IV: that explicitly estimating an evolving latent patient state yields more temporally consistent, better-calibrated, and more clinically faithful longitudinal reasoning than treating history as contextual input. CBS is evaluated as the concrete implementation under test, but the experiments are designed to probe the underlying formulation—temporal consistency, contradiction handling, uncertainty calibration, and robustness to missing longitudinal context—rather than to establish that CBS is simply “better” than prior longitudinal VLMs in an aggregate sense. Evaluation therefore assesses both report quality and the behavior of the estimated patient state.

B. Datasets

1) *MIMIC-CXR*: The primary evaluation dataset is the MIMIC-CXR longitudinal collection, offering a large-scale patient population, paired chest radiographs and reports, longitudinal follow-up, realistic clinical variability, and public availability. Patient trajectories are reconstructed chronologically, with all studies from a single patient kept within the same data split to prevent leakage.

We report the dataset’s longitudinal density explicitly rather than leaving it implicit, since it directly bounds what the stress tests in Section VI-I can support: by patient visit count, MIMIC-CXR contains approximately 33,922 patients with exactly one study (no longitudinal signal), 10,490 with two, 5,079 with three, 3,021 with four, 1,968 with five, and 6,067 with more than five. Only the roughly 44% of patients (approximately 26,625) with two or more studies support any longitudinal modeling, and the long-horizon / catastrophic-forgetting stress test in particular has a realistically usable pool

of approximately 6,000–8,000 patients (those with five or more visits). We scope the long-horizon stress test to this subset explicitly and note that it may be biased toward chronically ill patients, limiting generalizability of long-horizon findings specifically (Section VI-M).

2) *Open-I*: Open-I is used for external validation to evaluate generalization. Given its substantially smaller scale relative to MIMIC-CXR, we treat it as a qualitative external-validation check rather than a second quantitative benchmark for the stress tests, unless a dataset audit at implementation time confirms sufficient sample size for a given stress-test condition.

3) *MI-CXR*: We additionally evaluate on MI-CXR, a benchmark of five-visit CXR timelines constructed specifically to test temporal event localization, interval-wise change reasoning, and global trajectory summarization [23]. MI-CXR reports that fourteen current state-of-the-art VLMs score only approximately 29% on its tasks, barely above the 20% chance rate for its five-way questions, which both motivates this work and provides a second, purpose-built quantitative benchmark for longitudinal reasoning beyond MIMIC-CXR and Open-I.

4) *Future Extensions*: The framework is modality-independent and could extend to CheXpert, PadChest, longitudinal CT datasets, and multimodal electronic health records.

C. Data Preprocessing

Each patient trajectory is sorted chronologically. Preprocessing includes image normalization, report tokenization, meta-data extraction, temporal interval computation, and removal of incomplete trajectories when necessary. Future reports are never available during training or inference.

D. Baseline Models

- **Baseline 1 – Cross-sectional VLM**: uses only the current radiograph; evaluates the benefit of longitudinal reasoning.
- **Baseline 2 – Previous Report Prompting**: appends the previous report to the language model prompt; evaluates textual history conditioning.
- **Baseline 3 – Multi-image Longitudinal Encoder**: jointly encodes current and previous radiographs; evaluates image-based temporal context.
- **Baseline 4 – CXRMate**: prompt-conditioned longitudinal report generation [1].
- **Baseline 5 – CXRMate-2**: state-of-the-art longitudinal reporting based on retrieval and multimodal re-encoding, integrating structured multimodal conditioning with a reinforcement-learning composite reward for semantic alignment [2]; repository inspection indicates historical studies are reconstructed at each inference rather than represented by a persistent patient state, making this the most appropriate architectural comparison for CBS. We note that CXRMate-2’s own published evaluation already includes a blinded, randomized retrospective radiologist read; our planned human evaluation (Section VI-K) should accordingly be framed as applying this evaluation design to a state-estimation model under stress-test

conditions, not as introducing a novel evaluation design of its own.

- **Baseline 6a – MAIRA-2 (reimplemented, data-matched):** MAIRA-2’s checkpoint (RAD-DINO-MAIRA-2 + Vicuna-7B) is publicly released under gated access [17], but was trained on a mixture of public and private data, so evaluating it zero-shot would not be data-matched against a CBS trained only on MIMIC-CXR. Our *primary* comparison is therefore a reimplementation of MAIRA-2’s grounded-conditioning architecture, trained on MIMIC-CXR only, reported alongside CBS and all other baselines in every main results table.
- **Baseline 6b – MAIRA-2 (public checkpoint, unmatched reference):** the released public checkpoint, evaluated zero-shot without further training, reported separately and clearly labeled as an unmatched, production-scale reference point rather than a data-matched comparison; it is excluded from the primary capacity-matched and compute-normalized comparisons (Section VI-D) for this reason.
- **Baseline 7 – MLRG-style missingness handling:** a context-conditioned baseline using tokenized-absence encoding for missing priors [20], included to isolate whether any CBS advantage under missing-history stress tests (Section VI-I) is attributable to explicit state estimation specifically, rather than to better missingness handling in general, which MLRG addresses without recursive state estimation.

All baselines and CBS will be matched on backbone vision encoder and language model where feasible, and matched on total trainable parameter count as closely as architecturally possible, for every baseline except the explicitly unmatched Baseline 6b; any residual parameter-count gap introduced by CBS’s additional recurrent and variational components (GRU/SSM memory, prior and posterior networks) will be reported explicitly, and results will additionally be reported as a compute-normalized comparison (accuracy per GPU-hour, Section VI-L) so that reviewers can assess whether observed gains survive a capacity-matched comparison rather than reflecting CBS simply having more parameters (Section VI-M).

E. Training Protocol

Patients are divided into training, validation, and test sets using patient-level splits. Training proceeds chronologically through each patient trajectory using AdamW, mixed precision training, early stopping, and cosine learning-rate scheduling, with hyperparameters selected via validation performance.

F. Evaluation Metrics

1) *Report Generation:* BLEU, ROUGE-L, METEOR, CIDEr, together with RadCliQ [24], a regression-based composite metric shown to track radiologist-identified error counts more closely than any single n-gram metric, and GREEN [25], an LLM-based, error-typed metric with interpretable explanations; RaTEScore and FineRadScore are reported as optional

supplementary metrics where compute allows. We include RadCliQ and GREEN because the 2020–2023 metric set alone is no longer considered sufficient evidence of report quality in this literature.

2) *Clinical Accuracy:* CheXbert F1, RadGraph F1, entity overlap, finding accuracy, impression accuracy.

3) *Longitudinal Reasoning:*

- *Temporal Consistency* – whether reports evolve consistently across sequential examinations, without unsupported oscillation between resolved and present states.
- *Disease Progression Accuracy* – whether improvement, worsening, or stability are correctly described, scored against CheXTemporal’s five-class progression taxonomy (new, resolved, improving, worsening, stable) rather than an unlabeled notion of progression.
- *Contradiction Detection* – whether the model appropriately revises previous beliefs given contradictory observations.
- *Belief Persistence* – how much clinically relevant information is retained across successive encounters.
- *Longitudinal State Tracking* – whether the estimated latent state’s implied progression agrees with CheXTemporal’s spatial-temporally aligned progression labels (Section III-G) over time; this replaces the previously unspecified “known disease trajectories” ground truth with an operational, citable label source, and is only reported for the subset of trajectories with CheXTemporal coverage.

4) *Uncertainty Evaluation:* Expected Calibration Error (ECE), Negative Log Likelihood (NLL), Brier Score, predictive entropy.

G. Latent State Evaluation

- *Trajectory Visualization:* latent states Z_t projected via UMAP, t-SNE, and PCA for qualitative inspection only; visualization is treated as illustrative and is not used on its own as evidence of disease-specific structure (Section IV-F).
- *Quantitative Clustering:* silhouette score and k NN purity of the latent space against CheXTemporal- and CheXpert-derived disease labels, computed across independently trained seeds and reported with confidence intervals, replacing visualization-only claims as the primary evidence for disease-specific latent structure.
- *Temporal Smoothness:* the distance $\|Z_t - Z_{t-1}\|$ is analyzed; healthy progression should produce gradual changes, acute disease larger transitions.
- *Belief Drift:* longitudinal drift is analyzed over extended patient histories (the ≥ 5 -visit subset identified in Section VI-B) to assess stability and resistance to catastrophic forgetting.
- *Causal Perturbation Probe:* to address the recurrent-capacity-vs-semantics confound (Section VI-Q) with evidence beyond correlational clustering, we counterfactually perturb a single component of O_t at a given encounter (e.g., swap in an image or finding consistent

with a different disease state, holding the rest of O_t fixed) and measure whether the resulting change in Z_t is concentrated in the latent dimensions identified as disease-relevant by the clustering analysis above, versus diffuse across the latent space in a manner more consistent with generic sequence-capacity effects.

H. Ablation Studies

Ablations D and E together isolate the confound named in Section I-D and Section VI-Q most directly: that improvements attributed to CBS could reflect generic recurrent-modeling capacity rather than patient-state semantics specifically. We make this isolation explicit as a five-condition matrix, run in Phase 1 (Section VI-M) before any large-scale decoder training, rather than leaving the comparison implicit across separately listed ablations:

TABLE II
CONFOUND-ISOLATION MATRIX (PHASE 1, FROZEN BACKBONE)

Condition	History	Recurrence	Stochastic
A. Cross-sectional	current only	No	No
B. Retrieval	retrieved	No	No
C. Recurrent memory	full	Yes	No
D. CBS	full	Yes	Yes
E. Context-transformer	full	Transformer	No

The key comparison is D vs. C: if CBS (D) does not outperform the generic recurrent-memory baseline (C) under the stress-test conditions (Section VI-I), the “belief-state” interpretation is weak regardless of whether CBS outperforms A, B, or E, and we treat D vs. C, not aggregate performance against all baselines, as the primary test of the patient-state-semantics claim specifically. Condition E additionally guards against the objection in Section VIII-A (long-context transformers as an alternative to recurrence entirely).

The remaining ablations refine this comparison:

- Remove deterministic memory (stochastic belief only), within the full CBS architecture.
- Remove stochastic latent belief (deterministic memory only); together with the matrix above, this isolates the specific contribution of H_t vs. Z_t within CBS, distinct from the C-vs-D comparison, which isolates CBS as a whole against generic recurrence.
- Remove KL regularization to evaluate uncertainty modeling, and, if $\mathcal{L}_{\text{consistency}}$ (Section IV-H) is found redundant with the standard KL term, drop it here and report the comparison.
- Remove previous reports to evaluate dependence on textual history.
- Replace recursive belief update with history retrieval, directly comparing CBS against retrieval-and-reencoding (equivalent to matrix condition D vs. B above, at full architectural scale). This experiment most directly tests the central hypothesis of the proposal and will be run first, as a small-scale pilot (Section VI-M), before committing to full-scale training.

- Vary latent dimension to evaluate belief capacity.
- Compare recurrent architectures: GRU, Mamba, and State Space Models.
- Frozen-decoder / soft-prompt conditioning: freeze the pretrained LLM decoder and inject (H_t, Z_t) as a soft-prompt conditioning prefix, training only the encoder and recurrent state modules, instead of backpropagating through the full decoder via truncated BPTT. We revise this from a fallback to the *primary* implementation strategy for Phases 1–3 (Section VI-M): the confound-isolation matrix above and the stress tests in Section VI-I are run entirely under frozen-decoder conditioning first, since the scientific question (does explicit state estimation help, holding capacity fixed) does not require a fully fine-tuned 7B+ decoder to answer. Full end-to-end BPTT-through-the-LLM (Phase 4) is attempted only after Phases 1–3 establish that CBS outperforms condition C under frozen-decoder conditioning; if it does not, we do not proceed to the more expensive full fine-tuning regime, consistent with the falsifiability commitment in Section I-D.

I. Stress-Test Evaluation

Because the central hypothesis of this work concerns behavior under partial observability, we introduce a dedicated set of stress tests designed to directly probe the conditions under which patient-state estimation is expected to matter most, rather than leaving such conditions scattered across the metrics above. For each condition, a quantitative pass/fail threshold constituting support for the hypothesis will be pre-registered before the full-scale experiments are run. We commit now to the *method* for setting these thresholds—a minimum clinically meaningful effect size relative to a capacity-matched baseline, calibrated against a fixed statistical-power target (Section VI-O)—rather than choosing exact numeric values only after seeing pilot results, so that threshold-setting cannot function as post hoc threshold-shopping; the pilot (Section VI-M) supplies the variance estimate the power analysis requires, and the resulting numeric thresholds are reported in the completed paper regardless of outcome.

- Missing prior studies:* one or more historical encounters are withheld at inference time, evaluating whether belief persistence degrades gracefully.
- Contradictory follow-up reports:* we prioritize naturally occurring contradictions mined from inter-radiologist disagreement subsets and genuine disease-course ambiguity where identifiable in MIMIC-CXR. A concrete candidate source is a small, multi-radiologist-annotated subset in the style of ReXVal (used in the RadCliQ/GREEN validation literature), which we will check for availability and scale early in the pilot phase (Section VI-M) rather than assuming it exists at sufficient size; synthetic contradiction injection (adversarially inserted negations) is used only as a secondary, explicitly labeled proxy condition if the naturally occurring subset proves too

small, since manufactured contradictions may not reflect the distribution of real clinical disagreement.

- *Long-horizon degradation*: performance and calibration are tracked as patient history length grows, scoped to the ≥ 5 -visit subset (Section VI-B), evaluating belief drift and resistance to catastrophic forgetting.
- *Irregular sampling*: trajectories with highly variable inter-visit intervals Δt are evaluated separately from regularly sampled trajectories.
- *Counterfactual history masking*: portions of a patient’s history are systematically masked or permuted to test whether belief updates respond appropriately to the available evidence rather than to spurious positional or ordering cues.

These conditions are the primary empirical basis for the falsifiability criterion stated in Section I-D: if CBS does not outperform retrieval- and context-conditioned baselines under these stress tests specifically, against the pre-registered thresholds above, this would count as evidence against the patient-state estimation hypothesis, independent of aggregate performance on standard, non-adversarial splits. We commit to reporting null or mixed results honestly if the pre-registered thresholds are not met, rather than revising the framing post hoc to fit the observed outcome.

J. Statistical Analysis

Performance differences are evaluated using paired t-tests, Wilcoxon signed-rank tests, and bootstrap confidence intervals, with results reported using 95% confidence intervals. The number of hypothesis tests to be run will be fixed in advance (one per stress-test condition per primary metric, plus the ablation comparisons in Section VI-H), so that Holm–Bonferroni or Benjamini–Hochberg correction can be selected and applied a priori rather than chosen post hoc based on which comparisons turn out significant.

K. Qualitative Evaluation

Representative patient trajectories (e.g., improving pneumonia, worsening edema, stable fibrosis, recurrent pleural effusion) are examined by comparing the ground-truth report, baseline prediction, CBS prediction, and belief trajectory visualization. Human evaluation will use at least three board-certified or senior-resident radiologists, blinded to model identity and presented with outputs in randomized order, assessing whether generated reports correctly reflect disease evolution, maintain consistency with prior examinations, and appropriately incorporate new findings; inter-rater reliability will be reported using Fleiss’ or Cohen’s kappa as appropriate. Radiologist collaborators for this component will be identified and confirmed before Month 1 of the schedule in Table ??, given the resource- and access-dependence of this evaluation. If recruitment slips beyond this point, we will proceed with the automated-metric evaluation (Section VI-F, including RadCliQ and GREEN) as the evaluation floor and treat the human read as a stretch goal to be added opportunistically, rather than

allowing the entire evaluation program to block on radiologist availability.

L. Computational Evaluation

Runtime experiments compare training time, inference latency, GPU memory usage, parameter count, and scalability with patient history length, reported as the compute-normalized comparison (accuracy per GPU-hour) introduced in Section VI-D. Because CBS stores a fixed-dimensional latent state, it is hypothesized to reduce dependence on repeatedly processing long histories; the extent of any efficiency gains will be determined empirically.

We budget this explicitly rather than leaving it unstated, and revise the budget downward relative to earlier drafts by making frozen-decoder conditioning (Ablation H) the primary strategy for Phases 1–3 (Section VI-M) rather than a fallback. Under frozen-decoder conditioning, only the encoder, recurrent state, and a lightweight adapter or LoRA projection are trained, which is feasible on a single high-end GPU or small multi-GPU node and does not require truncated BPTT through a 7B+ decoder. Full end-to-end fine-tuning of a LLaMA-family medical LLM (7B or larger), conditioned on a per-encounter recurrent state and trained with truncated backpropagation through time across patient trajectories at MIMIC-CXR scale, is deferred to Phase 4 and attempted only if Phases 1–3 support the hypothesis; if attempted, it is expected to require multi-GPU training ($4\text{--}8 \times \text{A100/H100}$ -class accelerators) over a period of weeks rather than days, and remains a named engineering risk for that phase specifically, not a precondition for the earlier, scientifically decisive phases. A realistic end-to-end timeline for a single strong PhD-student-scale effort is 9–14 months; we adopt the upper end, 14 months, as our planning baseline (Table ??) to avoid underbudgeting, but note that this budget is now substantially de-risked relative to a design that required Phase 4-level compute from the outset.

M. Phased Implementation Plan and Pilot

Rather than committing to full-scale, full-decoder training as the first experiment, we adopt a four-phase plan in which each phase is a decision gate for the next, with the transition from Phase 1 to Phase 2 constituting the primary pilot referenced elsewhere in this proposal (Sections I-D, VI-I, VI-O).

Phase 1 – Establish the phenomenon (frozen backbone, ~1,000–2,000 MIMIC-CXR patients). Run the confound-isolation matrix (Table II, conditions A–E) under frozen-decoder conditioning (Ablation H). This is the pilot: if CBS (condition D) does not outperform the generic recurrent-memory baseline (condition C) on the stress-test conditions at pilot scale, this is treated as a signal to revise the framing or implementation before further writing or compute is committed, not as a result to discard in favor of proceeding regardless. Pilot results, including a null or mixed outcome, will be reported in the completed paper alongside full-scale results.

Phase 2 – Stress test. Using the Phase 1 models, apply the full stress-test battery (Section VI-I: missing history,

contradictory history, long history, irregular intervals, counterfactual masking) and compute $\Delta = \text{Performance}(\text{CBS}) - \text{Performance}(\text{Retrieval})$ under each condition, against the pre-registered thresholds from the power analysis (Section VI-O).

Phase 3 – Inspect the state. Test whether changing clinically relevant evidence causes predictable, localized changes in Z_t using the causal perturbation probe (Section VI-G) and prediction probes on constructed evidence sequences (e.g., no effusion $\rightarrow$ small $\rightarrow$ moderate $\rightarrow$ resolved), rather than relying on UMAP/t-SNE visualization alone.

Phase 4 – Scale, only if Phases 1–3 support the hypothesis. Scale to the full decoder (7B+, end-to-end BPTT or LoRA fine-tuning), run the complete MIMIC-CXR and MI-CXR evaluation, and add the radiologist human evaluation (Section VI-K). If Phases 1–3 do not support the hypothesis, we do not proceed to Phase 4, and report the Phase 1–3 findings as the primary contribution instead; a rigorous negative result under this framing is still a valid scientific contribution (Section VI-O, Section I-D falsifiability commitment), and is treated as such rather than as a failure to be avoided by quietly abandoning the pre-registered comparison.

N. Primary and Secondary Outcomes

The full evaluation program in this section (Sections VI-F–K) is broader than any single thesis chapter needs to resolve, and we prioritize it explicitly rather than treating every metric as equally load-bearing. We designate three primary outcomes, prioritized as pre-registered comparisons under the power analysis (Section VI-O); all other metrics in this section are secondary or exploratory and are reported but not treated as individually decisive:

- P1. Longitudinal reasoning:** does CBS improve Disease Progression Accuracy (Section VI-F.3) over retrieval and context-conditioned baselines?
- P2. Missing-history robustness:** how does performance degrade as portions of the history $E_{1:t-k}$ are withheld (Section VI-I), relative to retrieval and context-conditioned baselines?
- P3. Belief calibration:** does the stochastic component Z_t provide better-calibrated predictions of a clinically defined target (Section VI-F.4) than post-hoc uncertainty-quantification baselines (Section III-H)?

This prioritization is what the Phase 1–2 pilot (Section VI-M) is designed to answer first, and is what determines whether Phase 4 is attempted at all.

O. Statistical Power at Stress-Test Scale

The long-horizon stress-test subset (≥ 5 visits, approximately 6,000–8,000 patients, Section VI-B) is further divided across train/val/test splits and across ablation and baseline comparisons; after a generous 70/15/15 split this leaves on the order of 900–1,200 test patients, and further stratification by contradiction status, irregular-sampling status, or missingness condition for the other stress tests (which likely draw from overlapping patients) could leave individual stress-test

cells with only low hundreds of patients, before the Holm–Bonferroni or Benjamini–Hochberg correction across metrics, stress conditions, and ablations (Section VI-J) further tightens the effective significance bar. Before committing to Phase 4, we will conduct a minimum-detectable-effect-size analysis using the Phase 1–2 pilot’s observed variance and the planned multiple-comparison correction. If the realistically available per-cell N cannot detect the smallest clinically meaningful effect size at 80% power, we will consolidate the stress-test battery to the three primary outcomes above (Section VI-N), reporting the remainder as secondary or exploratory, rather than treating an underpowered null result as evidence against the hypothesis. This power analysis, not the pilot’s point estimates alone, is what determines the final pre-registered thresholds referenced in Section VI-I.

P. Expected Outcomes

These experiments are designed to validate the patient-state estimation formulation, not merely to rank CBS against prior architectures. The framework is expected to demonstrate improvements in longitudinal consistency, disease progression modeling, uncertainty calibration, latent trajectory coherence, robustness to missing longitudinal context, and clinically meaningful report generation. Even if improvements in traditional language metrics are modest, evidence that CBS produces more stable, better-calibrated, and more interpretable patient-state representations would support the central hypothesis that explicit state estimation captures aspects of longitudinal clinical reasoning that context concatenation or retrieval alone do not.

Q. Threats to Validity

Internal Validity: performance differences may arise from architectural capacity rather than CBS itself; backbones will be matched across baselines where practical. **Construct Validity:** no established benchmark exists for latent patient belief representations; proposed belief persistence and trajectory analyses are exploratory. **External Validity:** experiments focus on chest radiography; generalization to other modalities remains future work. **Dataset Limitations:** radiology reports reflect clinical interpretation rather than absolute ground truth. **Confound of Recurrent Dynamics vs. Patient-State Semantics:** improvements attributed to CBS may arise from recurrent latent dynamics in general (e.g., any GRU- or SSM-based sequence model) rather than from patient-state semantics specifically. Ablations E and G, together with the stress-test evaluation in Section VI-I, are designed to help disentangle these two explanations, but a residual confound cannot be fully ruled out with observational comparisons alone.

VII. EXPECTED CONTRIBUTIONS

This work makes four contributions, listed in order of scientific priority. The first is the primary contribution; the remaining three follow from it.

Contribution 1. A principled and empirically testable computational formulation of longitudinal radiology as sequential patient-state estimation under partial observability,

distinguishing the hidden patient state s_t , the recursive posterior $p(s_t | E_{1:t})$, and the downstream report-generation task $p(r_t | s_t, E_{1:t})$.

Contribution 2. A structured latent dynamical implementation (CBS) that operationalizes this formulation, decomposing s_t into a deterministic disease trajectory memory H_t and a stochastic latent component Z_t , updated recursively via prior prediction and posterior correction in the RSSM tradition; whether Z_t ’s variance is calibrated against clinically defined uncertainty, rather than merely useful residual variability for report prediction, is treated as an empirical question (Section VI-F.4) and not assumed by construction.

Contribution 3. Longitudinal multimodal report generation through recursive patient-state updates instead of history concatenation, retrieval, or re-encoding, propagating a fixed-dimensional belief (H_t, Z_t) between encounters rather than reconstructing historical context at every inference call.

Contribution 4. Experimental evidence, obtained by testing the patient-state estimation hypothesis directly (Section VI), that explicit patient-state estimation improves temporal consistency, contradiction handling, and uncertainty calibration relative to context-concatenation and retrieval-based baselines.

Based on our literature review and repository analyses conducted for this proposal, this formulation—and its instantiation for multimodal longitudinal chest radiography—has not been previously proposed. Put plainly: our contribution is primarily an empirical investigation of a computational formulation, not a new probabilistic inference algorithm; CBS is presented as the concrete implementation of that formulation.

VIII. DISCUSSION

Reframing longitudinal radiology as patient-state estimation changes what counts as evidence for or against a given system. Rather than asking only whether a model produces higher-scoring reports, we ask whether it produces an estimate of the patient’s condition that is persistent, uncertainty-aware, and appropriately revised in light of new evidence.

Four aspects of the formulation deserve particular emphasis. First, *partial observability*: no single encounter fully reveals the patient’s underlying condition, so any adequate account of longitudinal reasoning must treat the patient state as inferred rather than directly given. Second, *uncertainty*: because the patient state is never fully observed, a faithful estimate must carry calibrated uncertainty rather than a single deterministic point estimate. Third, *recursive evidence accumulation*: each new encounter should update, rather than replace, the belief formed from prior encounters, which is the property that most clearly distinguishes state estimation from context concatenation. Fourth, *persistent patient state*: the belief must be carried forward as a compact computational object between encounters, rather than reconstructed from raw historical inputs each time.

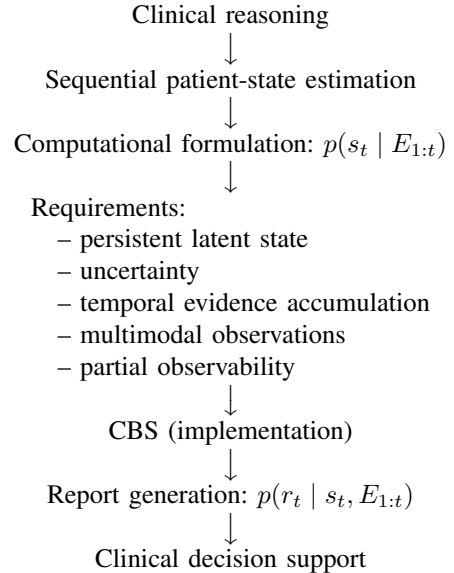
CBS operationalizes these four requirements through a structured latent dynamical model, but it does not introduce a new belief representation in the sense of a new theoretical construct; rather, it operationalizes the patient-state estimation

formulation using existing, well-established recurrent state-space modeling machinery, adapted to multimodal clinical observations and clinical state semantics.

A. Why Compression Over Long-Context Retrieval?

A natural objection is that, given the long context windows of modern LLMs and the demonstrated success of context-conditioned systems such as MAIRA-2 and Harrison.Rad at scale, explicit state compression may simply be unnecessary: why not condition on more raw history directly? We do not contest that context-conditioning can achieve strong aggregate report quality—indeed, we do not attempt to outperform proprietary, million-scale systems on that axis (Section I-D). Our claim is narrower and, we argue, orthogonal to context length: a fixed-dimensional belief with an explicit stochastic component provides (i) calibrated, trainable uncertainty over the patient’s underlying condition, which token-level generation confidence does not provide regardless of context length (Section III-H); (ii) a compact object whose trajectory can be probed, visualized, and, in principle, causally intervened upon independent of decoder behavior (Section IV-F); and (iii) explicit, structural handling of missingness and contradiction via the posterior network rather than implicit handling via attention over a longer prompt (Section IV-C.1). Long-context conditioning and explicit state estimation are not mutually exclusive; the empirical question we test (Section VI, Ablation E) is whether the latter provides measurable advantages specifically under the partial-observability and contradiction conditions where these three properties should matter most, not whether it is a wholesale replacement for retrieval-based conditioning at scale.

The relationship between the clinical motivation, the computational formulation, and the CBS instantiation is summarized below.



This pipeline makes explicit that CBS is an implementation of the formulation rather than the formulation itself: a different structured latent dynamical model satisfying the

same requirements—persistence, uncertainty, recursive evidence accumulation, multimodal observations, and partial observability—would equally constitute a valid instantiation of the same computational formulation, and would be expected to exhibit similar qualitative behavior on the longitudinal-reasoning evaluations proposed in Section VI.

IX. ETHICAL CONSIDERATIONS

CBS is intended to support radiologists rather than replace clinical judgment. All experiments will use de-identified publicly available datasets in accordance with their data use agreements and institutional requirements. Performance should be evaluated across demographic groups where dataset annotations permit such analysis. Where feasible, analyses of latent belief trajectories and uncertainty estimates should accompany quantitative performance results to improve interpretability. Any future clinical deployment would require prospective validation, regulatory review, and integration with established clinical governance processes.

X. LIMITATIONS

We explicitly acknowledge that CBS is not a new probabilistic modeling family. It belongs to the broader class of variational latent state-space models, and its recursive belief-update equations, prior–posterior decomposition, and KL-regularized training objective are adopted from the established RSSM literature rather than introduced by this work. The novelty claimed by this proposal lies in the problem formulation (longitudinal radiology as sequential patient-state estimation under partial observability), the structured clinical state semantics used to instantiate that formulation, the longitudinal multimodal integration required to apply it to chest radiography, and the empirical validation of the resulting hypothesis—not in the underlying probabilistic machinery itself.

Beyond this framing, several practical limitations remain. CBS requires longitudinal patient trajectories and cannot exploit recursive belief updating on single-image datasets. Latent variables remain indirect abstractions requiring additional interpretability methods. Compared with standard report-generation models, CBS introduces recurrent computation, latent inference, and variational optimization, likely increasing computational requirements. Uncertainty calibration and clinical validation remain open challenges requiring prospective, physician-in-the-loop studies.

XI. PROJECT TIMELINE

We reconcile the 9–14 month estimate in Section VI-L with a single, revised 14-month schedule (rather than the twelve months given in earlier drafts), to reflect the anticipated cost of BPTT-through-LLM debugging identified there. The pilot study (Section VI-M) and the power analysis it feeds (Section VI-O) are scheduled explicitly within Months 4–5, before full-scale training begins, so that their outcomes can inform whether the full 14-month plan or the reduced-scope fallback (Ablation H, frozen-decoder conditioning) should proceed.

XII. RISK ASSESSMENT

TABLE III
RISKS AND MITIGATIONS

Risk	Mitigation
Limited performance improvement	Systematic ablation studies isolating CBS contribution
Training instability	Staged training, gradient clipping, hyperparameter tuning
Posterior collapse	KL annealing or free-bits strategies
Computational cost	Begin with smaller latent dimensions, scale progressively
Limited interpretability	Complement quantitative results with visualization and probing
BPTT-through-LLM instability	Named fallback: frozen-decoder / soft-prompt conditioning (Ablation H), not just a mitigation note
Radiologist recruitment slips	Automated-metric-only evaluation (RadCliQ, GREEN) as the floor; human read as stretch goal
Underpowered stress-test cells	Pre-registered power analysis (Section VI-O); consolidate to 2–3 primary comparisons if needed

XIII. FUTURE RESEARCH DIRECTIONS

Future work may incorporate additional modalities (laboratory values, medications, clinical notes, ECG, genomics) into a multi-modal Clinical Belief State; explore continuous-time disease modeling using neural ordinary differential equations or continuous-time state-space models; develop explainable clinical belief states that associate latent dimensions with clinically interpretable concepts; extend toward foundation clinical world models spanning imaging, text, laboratory data, and interventions; and investigate belief trajectories for prognosis, disease forecasting, treatment planning, and risk prediction, provided appropriate task-specific datasets and prospective validation are available.

XIV. CONCLUSION

This paper has proposed sequential patient-state estimation under partial observability as a computational formulation for longitudinal radiology. Existing methods largely treat history as contextual input for report generation; we instead formulate longitudinal reasoning as recursive inference over an evolving latent patient state, and instantiate this formulation using a structured variational recurrent state-space model, the Clinical Belief State (CBS). The primary contribution of this work is the computational formulation and its empirical validation; CBS is the concrete implementation of that formulation, combining a deterministic disease trajectory memory with a stochastic, uncertainty-aware clinical belief to integrate multimodal observations over time, revise beliefs in response to new evidence, and condition report generation on an evolving estimate of patient health. We do not claim this is the only viable formulation of longitudinal reasoning, nor that CBS is the only viable implementation of it; rather, we present a falsifiable

hypothesis and an evaluation protocol, including explicit stress tests, designed to determine whether this formulation offers measurable advantages over retrieval- and context-conditioned alternatives. If supported experimentally, this repositioning—from history-aware report generation to belief-aware patient-state estimation—would offer a principled and testable foundation for longitudinal medical vision-language modeling.

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